

Section - I: Multiple Choice Questions

1. What is a common characteristic of a model that is underfitting the training data?
 - A) It performs well on the training data but poorly on new unseen data.
 - B) It is complex and has many parameters relative to the amount of training data.
 - C) It fails to capture the underlying patterns in the training data.
 - D) It is biased towards a particular class or output.

Section - II: Descriptive / Case Study-Based Questions

1. A data scientist, Emma, is building a machine learning model to predict house prices based on features such as the number of bedrooms, square footage, and location. She collects a dataset of 1000 houses and trains a model using 80% of the data for training and 20% for testing. After training the model, Emma observes that the model performs extremely well on the training data, with an R-squared value of 0.95, but poorly on the testing data, with an R-squared value of 0.5. Explain the likely problem with Emma's model and suggest one potential solution to address this issue.